

Thrust-Efficiency Case Study

Apollo LM Powered Descent with a Partial-Thrust Engine

30 July 2026

Contents

1	Thrust-Efficiency Case Study	1
1.1	The headline result	2
1.1.1	The three regimes	2
1.2	Results	3
1.2.1	Regime I is graceful; the penalty is propellant	4
1.2.2	Regime II converges but does not land acceptably	4
1.2.3	Regime III is infeasible	4
1.3	Relationship to the other two studies	4
1.4	What the model does and does not capture	5

1 Thrust-Efficiency Case Study

Engine 2 delivers only a fraction η of its commanded thrust as useful force,

$$F_{\text{delivered}} = \eta T, \quad \eta \in (0, 1],$$

while its actuator dynamics — and therefore its propellant draw — still act on the full T . The shortfall is **lost power, not saved fuel**. Engine 1 is healthy; everything else is held at the baseline. Implemented through the new `LMPparams.thrust_eff_eng`.

Case	Directory	η	Regime	Result
E1	E1_eta1.00/	1.00	baseline	lands, $J = 2.867\text{e}8$
E2	E2_eta0.85/	0.85	throttle-only trim	lands, +1.8 %
E3	E3_eta0.65/	0.65	throttle-only trim	lands, +12.0 %
E4	E4_eta0.40/	0.40	throttle-only trim	lands, +30.4 %
E5	E5_eta0.25/	0.25	gimbal + RCS committed	converges but misses the landing gate
E6	E6_eta0.15/	0.15	untrimmable	no solution

Each directory holds `states.png`, `controls.png`, `actuators.png`, `thrusters.png`, `trajectory_with_axes.png`, `console_log.txt`, `metrics.json` and `solution.npz`. Three cross-case figures sit at the top level:

- `efficiency_regime_map.png` — the analytic regime boundaries
- `efficiency_comparison.png` — the converged trajectories overlaid
- `efficiency_cost.png` — what the loss costs

Reproduce with:

```
python run_efficiency_study.py      # all six cases, about 12 min
python plot_regime_map.py          # analytic regime map (no solve)
python plot_efficiency_comparison.py # trajectory overlay + cost figures
```

1.1 The headline result

The binding constraint is not the lost thrust — it is the roll asymmetry the loss creates. Losing efficiency on one engine of a laterally-spaced pair is primarily a *roll-trim* problem, exactly as in the engine-out study.

Even at $\eta = 0.15$ the vehicle retains 25 900 N of total delivered thrust against a 12 530 N hover requirement — a thrust-to-weight of 2.07, ample on paper. It still cannot land, because the two engines sit at $y = \pm 1.5$ m and unequal *delivered* thrust is a roll moment.

1.1.1 The three regimes

Balanced hover requires both engines to deliver $T_{\text{hover}}/2$, so the weak one must be commanded at $(T_{\text{hover}}/2)/\eta$ — and that command is bounded by $T_{\text{max,eng}} = 22\,520$ N. This gives the first boundary:

$$\eta_{\text{sat}} = \frac{T_{\text{hover}}/2}{T_{\text{max,eng}}} = \frac{6265}{22\,520} = \mathbf{0.278}$$

Below η_{sat} the weak engine saturates, the delivered thrusts can no longer be equalised, and the leftover asymmetry becomes a residual roll moment $L = y_{\text{eng}}(F_1 - F_2)$ that the gimbal and RCS must absorb. Setting that equal to the available authority gives the second boundary:

$$\eta_{\text{min}} = \frac{T_{\text{hover}} - L_{\text{auth}}/y_{\text{eng}}}{2 T_{\text{max,eng}}} = \mathbf{0.198}, \quad L_{\text{auth}} = 3293 + 2140 = 5432 \text{ N} \cdot \text{m}$$

Regime	Condition	Behaviour
I	$\eta \geq 0.278$	Trimable by differential throttling alone; gimbal and RCS stay free for manoeuvring
II	$0.198 \leq \eta < 0.278$	Trimable only with the gimbal and RCS committed to static roll trim
III	$\eta < 0.198$	Residual roll moment exceeds all available authority — infeasible

The authority is evaluated **at hover**, which is deliberate: the gimbal’s roll capacity is $(\sum_i F_i)d_{z,eng}\tan\delta_{y,\max}$, so it scales with delivered thrust and is *weakest* at hover — which is precisely where the vehicle must end up. During the high-thrust braking phase the same solutions carry roll moments of 7 000–10 500 N · m, well above the hover budget, without difficulty.

1.2 Results

Metric	$\eta=1.00$	0.85	0.65	0.40	0.25	0.15
Status	solved	solved	solved	solved	solved	FAILED
Objective J	2.867e8	2.919e8	3.211e8	3.737e8	5.626e8	—
J vs baseline	—	+1.8 %	+12.0 %	+30.4 %	+96.2 %	—
Peak delivered thrust [N]	45 036	41 658	36 254	22 867	16 848	11 881
Mean weak-engine command [N]	10 757	11 411	13 340	18 697	22 101	22 517
Mean healthy command [N]	10 430	10 336	10 347	9 563	8 593	5 716
Impulse burnt [N · s]	1.716e6	1.762e6	1.919e6	2.289e6	2.486e6	—
Impulse delivered [N · s]	1.716e6	1.623e6	1.540e6	1.380e6	1.144e6	—
Wasted impulse [N · s]	0	1.39e5	3.78e5	9.09e5	1.34e6	—
Mean TVC roll moment [N · m]	697	1 630	2 916	3 445	4 601	—
Mean yaw gimbal [°]	1.22	1.63	3.13	3.95	5.75	—
RCS impulse [N · s]	91 068	97 353	107 900	147 380	274 533	—
Time to contact [s]	26	27	30	41	74	—
Terminal speed [m/s]	3.8e-6	5.9e-6	2.6e-6	6.0e-6	0.193	25.5
Terminal attitude error [°]	~0	~0	~0	~0	5.95	34.2

Metric	$\eta=1.00$	0.85	0.65	0.40	0.25	0.15
IPOPT iterations	168	208	199	192	204	400 (capped)

1.2.1 Regime I is graceful; the penalty is propellant

For $\eta \geq 0.40$ every case lands to full precision (terminal speed below 10^{-5} m/s). The optimiser simply commands the weak engine higher — the mean command rises from 10 757 N to 18 697 N — and equalises the *delivered* thrusts. Roll trim costs nothing extra because differential throttling handles it.

What it does cost is propellant. At $\eta = 0.40$ the vehicle burns $2.289\text{e6 N} \cdot \text{s}$ of impulse to deliver $1.380\text{e6 N} \cdot \text{s}$: **909 kN · s, roughly 40 %, is wasted**. The descent also slows — thrust-to-weight falls from 3.59 to 1.80, so contact takes 41 s rather than 26 s.

1.2.2 Regime II converges but does not land acceptably

$\eta = 0.25$ sits between the two boundaries, and the model behaves exactly as the regime analysis predicts, on three independent signals:

- the weak engine’s mean command is 22 101 N against a 22 520 N limit — **it is saturated essentially throughout**;
- the mean yaw-gimbal deflection is 5.75° of the 6° available — **96 % of the gimbal range is consumed holding static trim**;
- RCS impulse triples, from 91 068 to 274 533 N · s.

The trajectory carries a persistent $\sim 3\,300$ N delivered-thrust asymmetry and $\sim 4\,800$ N · m roll moment for the entire descent, visible as the flat shelves in `efficiency_comparison.png`.

The NLP converges, but **the landing criteria are not met**: touchdown attitude error is 5.95° against the 5° gate, with 0.193 m/s residual speed. With the gimbal saturated and the RCS committed, there is nothing left to null the roll with. Contact also takes 74 s of the fixed 80 s horizon, so the vehicle is nearly out of time as well.

The practical limit is therefore $\eta \approx 0.28$, not 0.198. The analytic boundary at 0.198 is where a solution stops *existing*; the boundary at 0.278 is where an *acceptable landing* stops existing. For design purposes the latter is the number that matters, and it coincides with η_{sat} — the point where differential throttling alone can no longer equalise the delivered thrusts.

1.2.3 Regime III is infeasible

At $\eta = 0.15$ the residual roll moment at hover is $8\,662$ N · m against $5\,432$ N · m of authority — a deficit of $3\,229$ N · m. IPOPT exits with `Maximum_Iterations_Exceeded` after the 400-iteration cap with a constraint violation of 0.393 still outstanding (the converged cases reach 10^{-6}). The final iterate is 34.2° off in attitude and moving at 25.5 m/s. As in the engine-out study, the figures in that directory are a diverging iterate plotted for diagnosis, **not a trajectory**.

1.3 Relationship to the other two studies

All three fault studies converge on the same conclusion about this vehicle:

Study	Fault	Failure mode
Engine-out	one engine dead	roll authority — infeasible at design spacing
Degraded gimbal	one slow actuator	none — graceful, $\sim 10\text{--}20\%$ in J
Thrust efficiency	one weak engine	roll authority — unacceptable below $\eta \approx 0.28$

The two thrust-related faults both fail through roll, and both fail at the lateral spacing $y_{\text{eng}} = 1.5$ m rather than through any shortage of thrust. The gimbal fault is benign precisely because it does not create an asymmetry — it only makes one actuator slower to respond.

This strengthens the engine-out study’s design recommendation: the 1.5 m spacing, chosen to give differential-throttle roll authority, is what converts *any* thrust asymmetry into a trim problem the vehicle cannot absorb. A smaller spacing would raise η_{sat} tolerance in exactly the way it restored engine-out survivability.

1.4 What the model does and does not capture

- **η is constant and known from $t = 0$.** The optimiser plans around it with full knowledge. A real efficiency loss would appear mid-descent, and the open-loop OCP cannot represent detection delay or the transient.
- **Constant mass.** This is the most consequential simplification here, and it cuts against the reported numbers. The wasted impulse at $\eta = 0.25$ is $1.34\text{e6 N} \cdot \text{s}$, comparable to the *entire* delivered impulse. At an I_{sp} of ~ 311 s that is several hundred kilograms of extra propellant on a 7 711 kg vehicle — so a mass-tracking model would show the vehicle getting lighter faster, and might well run out of propellant before it runs out of roll authority. **The propellant consequence of a partial-thrust engine is understated by this model.**
- **η applies to force only.** The loss is modelled as a pure force scaling: no change to the thrust lag, no change to the throttle envelope, no extra thermal or vibration effect. A real cause (injector fouling, chamber erosion, feed-pressure loss) would likely also shift the throttle limits and the actuator response.
- **Both gimbal axes and the moment arm are unchanged.** The weak engine still gimbals normally, and its reduced thrust reduces its gimbal *authority* proportionally — which is captured — but nothing else about its geometry changes.
- **The objective comparison inherits the nonconvexity caveat** documented in the gimbal study: single-digit-percent gaps in J are not resolvable without a multi-start. Here the effect sizes ($+1.8\%$ to $+96\%$) are large and monotone in η , so the trend is safe even though the $+1.8\%$ entry on its own is not.
- **Feedback.** As throughout, these are open-loop reference trajectories.